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SPACE



**Auriga Space raises \$6M
to shoot rockets off an
electromagnetic launch
track**

A photograph of a large, industrial-scale electromagnetic launcher. It features a series of white, cylindrical coils stacked vertically on a metal frame. In the foreground, there's a complex arrangement of metal plates and structural components, some with a cross-hatched pattern. The background shows a large, open industrial space with a high ceiling and structural beams.

While U.S.-based companies are launching more payloads into orbit than ever before, they're still limited by the inefficiency of conventional rockets. With rockets, most of the weight is in the propellant, not the payload, and a huge portion of it is burned simply battling Earth's gravity and atmosphere.

Auriga Space is aiming to change the launch game. Instead of a first-stage booster, the California-based startup is developing a launch track that will use electricity to power powerful magnets. Those magnets will accelerate a small rocket to over six times the speed of sound. The track's final segment rises on a steep ramp, enabling the rocket to exit at hypersonic speed and only ignite its engine in the final push to orbit. Beyond propellant savings, the entire ground-based architecture is entirely (and rapidly) reusable.

IMAGE CREDITS: [AURIGA SPACE](#)

“Less than 2% of the mass of the rocket is what gets into space,” Auriga founder and CEO Winnie Lai explained in a recent interview. “So our ultimate goal here is to make space launch more efficient, and by increasing efficiency, we believe we can bring down the cost, and we can also enable much more frequent launches.”

Electromagnetic launchers are not a new idea: Auriga's architecture is reminiscent to a rail gun or a maglev. But Lai argues that recent advancements in power electronics, especially being able to operate at higher voltages and higher powers,

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finally make the concept technologically and commercially viable.

The company is fueling its ambitions with a previously undisclosed \$4.6 million seed round that closed at the beginning of the year and \$1.4 million in new AFWERX and SpaceWERX contracts. The capital round was led by European firm OTB Ventures, with participation from Trucks Venture Capital and Seraphim Space. Auriga has raised \$12.2 million across VC and DOD grants to-date.

The final system architecture, including the tunnel length and rocket size, are still being finalized. Even a very long track will still impart high-G loads on the vehicle, which could limit the types of satellites it can carry. The company has performed some initial studies on the survivability of satellite components under high-G loads that indicate they can survive higher Gs than standard testing assumed, Lai said. Auriga is also considering the opportunity to do “custom work,” like adding some structure support so objects can withstand the higher Gs, depending on what the customer needs.

“If you look at munitions, you also look at missile launches, those experience very, very, very high Gs,” she said. “So we’re very confident there’s payloads up there that could survive our launch environments, but that’s still yet to be defined. If we want to bring down the Gs, then we make the launcher longer.”

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Auriga will go to market first with hypersonic ground testing: This most recent direct-to-phase II Small Business Innovation and Research grant from AFWERX (the company's second) will be used to commercialize an indoor, lab-scale track called Prometheus early next year, and an outdoor accelerator for full-scale hypersonic test articles called Thor. (The orbital launcher is dubbed Zeus.)

Customers tell Lai that the lack of affordable, on-demand test infrastructure is one of the main challenges in advancing hypersonic development. Prometheus and Thor aim to fill that gap and provide a market capability that is sorely lacking: customers will be able to perform multiple tests on the same test article under different flight conditions, at a lower cost and higher cadence than what's currently available. It can also perform a number of different tests, like weather testing, aerodynamic tests, and impact testing using the same underlying architecture.

The company is also trying to address the demand from the U.S. Space Force for "responsive" launch, or the ability to launch payloads to orbit with little to no advance notice. While there have been some impressive demonstrations of responsive launch in the past 12 months, with companies like Firefly Space launching a rocket with a 24-hour notice, Auriga wants to shrink that time to a matter of minutes.

“We call an Uber, and we expect the Uber to show up in a matter of minutes,” she said. “I think that should be the case for space as well.”

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Aria Alamalhodaei

Reporter, Space and Defense |

Aria Alamalhodaei covers the space and defense industries at TechCrunch. Previously, she covered the...

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Space may be the final frontier, but there's nothing "final" about developing new and more cost-efficient ways to get there. Whether you have your sights set on Mars, delivering payloads to the International Space Station (ISS) or blasting satellites into geosynchronous orbit, it all starts with launch systems.

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All payloads, such as satellites, spacecraft, astronauts or supplies destined for the ISS, require rockets with enough propellant to launch and get them safely to their intended destinations. But it's not that simple: the scale and type of the launch system varies depending on the destination (thank you, Captain Obvious).

To achieve orbit, a launch vehicle must hit a minimum velocity of 17,500 mph. Flying to the moon or to Mars requires a minimum velocity of 25,000 mph. The massive amount — and weight — of propellant required to meet those speeds means the launch vehicle's structural weight must be kept to the bare minimum. And that's without getting into

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early-stage space startups.

Most space folks are well aware of the incredible achievements of both SpaceX (it just successfully launched its [fourth crewed mission](#) for NASA) and Rocket Lab ([which plans to launch](#) two commercial Earth-observation satellites on a two-stage Electron rocket). But it also pays to keep an eye on the up-and-comers — and you'll meet them at TC Sessions: Space 2021.



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Here's a prime example. Don't miss "New Kids on the Launch Block," a panel discussion with [Max](#)

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SpaceX details Starship and Super Heavy in new website

Darrell
Etherington

— 2:46 PM PDT · September 30, 2019

After CEO Elon Musk shared new details about its in-development Starship spacecraft on Saturday, SpaceX has updated its website with a [new section dedicated to the fully reusable cargo and passenger vehicle](#). The new Starship website also provides a bunch of info about Super Heavy, the first-stage booster that will propel Starship to orbital altitudes and beyond.

Starship, once complete, will be “world’s most powerful launch vehicle,” according to SpaceX, with a cargo capacity of 100 metric tons (that’s over 220,000 lbs) to Earth orbit. With orbital refueling, it’ll also be able to take its freight — and passengers — to the Moon, Mars and beyond.

Per the new Starship site, the final vehicle will be 160 feet tall (without the booster) and 30 feet in diameter, with a propellant capacity of 1,200 metric tons of liquid methane and liquid oxygen. Payload, and crew depending on configuration, will occupy the top third of the rocket, while the bottom two-thirds will house the propellant and six Raptor engines, including three for atmospheric flight and three for propulsion in space. At the top of the rocket there are two actuated (meaning you can control their movement) fins that will move to orient the rocket for re-entry and landing. At the bottom, two large fins will also help produce drag, crucial for its controlled descent. Starship will be made of stainless steel, and one half of its surface will be covered in glass tiles to take the brunt of the worst of the heat upon atmospheric entry.

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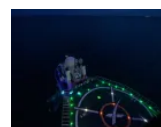
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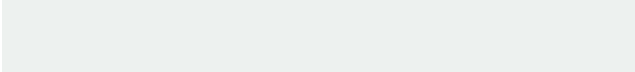
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As for Super Heavy, it'll have the same 30-foot diameter, but be much taller, at 223 feet, with a propellant capacity of a massive 3,300 metric tons, and a thrust capability of 72 meganewtons (MN) (the thrust of the Saturn V rocket was only 35 MN, by comparison). Actuated grid fins near the top of the rocket will be used for its controlled landing, much like those found on the Falcon 9 boosters SpaceX uses today. The bigger booster will have 37 Raptor engines, however, and six landing legs for stability when it comes back down to be readied for re-use. It'll be made entirely of stainless steel.

SpaceX notes that Starship is used to deliver satellites at “a lower marginal cost per launch” compared to Falcon, and that the design of its cargo compartment will provide the largest potential cargo fairing of any current planned space freight vehicle. This will allow it to carry even very large objects like assembled space telescopes, the company notes.

It's also designed to continue to service the International Space Station with cargo resupply, and would be able to transport a lot more cargo in one go than the current Dragon capsules SpaceX uses. It's also designed to be able to deliver cargo and people to the Moon and other planets, and to return for multiple trips.



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SpaceX also provided updated specs for the Raptor engine that will power the Starship system. Each Raptor will be 4 feet in diameter, 10.2 feet high and have a thrust capability of 2 MN (which is just over the thrust force of the Space Shuttle's main engine during takeoff, coincidentally).

On the site, SpaceX says it's targeting orbital flight for Starship in 2020, and at the event Musk went further still, saying that Starship should attain this milestone in less than six months — and even allowed for the possibility that it could fly with people on board in one year.

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